

Audrey E. Lee

audreliz.lee@gmail.com

September 15, 2026

<https://audreyelee.github.io/>

Education

Carnegie Mellon University (CMU)

Aug 2026 - May 2031

Ph.D. in Robotics

Relevant Courses: Human-Robot Interaction, Introduction to Robot Learning

Massachusetts Institute of Technology (MIT)

Aug 2025 - May 2026

M.Eng. in Electrical Engineering and Computer Science, Concentration in Artificial Intelligence

GPA: 5.0/5.0

Thesis: *Flexible Language-Guided Representation Editing for Adaptive Robot Behavior*

Relevant Courses: Algorithmic Human-Robot Interaction, Autonomy and Decision-Making, Intelligent Multimodal User Interfaces, Natural Language Processing

Massachusetts Institute of Technology (MIT)

Aug 2021 - May 2025

B.S. in Electrical Engineering and Computer Science, Minors in Mechanical Engineering and Writing

GPA: 4.9/5.0

Relevant Courses: Robotics - Science and Systems, Software Performance Engineering, Microcomputer Project Lab

Publications

- [1] Jocelyn Shen, **Audrey Lee**, Sharifa Alghowinem, River Adkins, Cynthia Breazeal & Hae Won Park. "Social Robots as Social Proxies for Fostering Connection and Empathy Towards Humanity". *IEEE/ACM International Conference on Human-Robot Interaction (HRI 2025)*
- [2] **Audrey Lee**, Nathan Dennler, Ilia Sucholutsky & Andreea Bobu. "Flexible Language-Guided Representation Editing for Adaptive Robot Behavior". *Conference on Robot Learning (CoRL 2026)*
- [3] Megan Wei, Melanie Subbiah, **Audrey Lee**, Annya Dahmani, Dave Edwards, Helen Edwards, & Ellie Pavlick. "Ownership in AI-Assisted Everyday Tasks". *In submission*
- [4] Annya Dahmani, Melanie Subbiah, Megan Wei, **Audrey Lee**. "Feedback and Credit Labels Reshape Ownership in AI-Assisted Work". *In submission*

Research Experience

CMU HARP Lab | Advisor: Henny Admoni

Aug 2026 - Present

- Exploring extension of robot learning systems that select human feedback modalities based on expected information gain to additionally account for real-time variation in feedback quality across user, task, and interaction context

MIT CLEAR Lab | Flexible Feature Editing Project (Advisor: Andreea Bobu)

Oct 2024 - Jul 2026

- Built a flexible feature-editing pipeline capable of adding, generalizing, or masking features to refine completeness, minimality, and interpretability of robot understanding from human natural language input and expert demonstrations
- Designed simulated, human-subject, and real robot experiments for first-author publication [2]
- **Publication:** *Flexible Language-Guided Representation Editing for Adaptive Robot Behavior*

MIT Robot Locomotion Group | Fast Path Planning Project (Advisor: Russ Tedrake)

Sep 2023 - May 2024

- Implemented real-time path planning using Drake through axis-aligned, collision-free bounding boxes
- Extended the approach to collision-free convex polytopes, analyzing tradeoffs in speed and generalizability
- Produced a poster and research paper for the MIT "Preparation for Undergraduate Research" class-internal poster session

MIT Personal Robots Group | Social Robots as Social Proxies Project (Advisor: Cynthia Breazeal)

Jun 2023 - Oct 2023

- Implemented a 5-phase interaction with AI-generated dialogue guiding participants to share stories with a Jibo robot over a 4-week study on embodied social agents (e.g., Jibo) as proxies for fostering human-to-human connection
- Built an integrated research platform to collect audio, video, real-time transcripts, and surveys using Firebase and Docker
- Coordinated a 40+ participant deployment, including research station packaging, onboarding, and remote technical support
- Performed quantitative survey analysis and qualitative interview coding, contributing to a peer-reviewed publication [1]
- **Publication:** *Social Robots as Social Proxies for Fostering Connection and Empathy Towards Humanity*

MIT Distributed Robotics Lab | *ActionSense Project (Advisor: Daniela Rus)* Sep 2022 - Jun 2023

- Developed a data processing framework for temporal alignment, visualization, and compact HDF5-based storage of 12+ hours of motion, tactile, and video sensing data streams from the ActionSense multimodal kitchen-task dataset
- Designed NN-based models to predict sensor readings for identifying redundant modalities and inter-sensor relationships

MIT Distributed Robotics Lab | *Air Guardian Project (Advisor: Daniela Rus)* Oct 2021 - May 2022

- Trained virtual autonomous aerial vehicles to navigate Unity environments using the RLlib Actor-Critic Agent Algorithm

Teaching

MIT 6.1903/4 Introduction to Low-level Programming in C and Assembly

Graduate Teaching Assistant (*30-person staff for 200+ students under Dr. Silvina Hanono-Wachman*) Spring 2026

- Led 2 weekly discussions covering C/Assembly syntax, stack and memory management, and programming representations
- Authored discussion and review session materials; staffed 10+ office hours/week with an average student rating of 6.9/7

Undergraduate Lab Assistant (*30-person staff for 200+ students under Dr. Silvina Hanono-Wachman*) Spring 2024, Spring 2025

- Provided 7.5+ hours/week of 1-on-1 support for C and RISC-V labs (debugging, memory management, and data storage)

MIT 6.4130/2 Principles of Autonomy and Decision Making

Graduate Teaching Assistant (*sole TA for 40+ students under Prof. Andreea Bobu*) Fall 2025

- Led weekly discussions to reinforce concepts in search, MDPs, reinforcement learning, Bayesian inference, and HMMs
- Authored practice worksheets, testing/visualization suites for 13 Python assignments, and course logistics documentation
- Hosted two weekly office hours and maintained an average < 30-minute response time on the online questions forum

MIT 6.3900 Introduction to Machine Learning

Tutor (*3 students per semester*) Fall 2024, Spring 2025

- Conducted individualized weekly review sessions and created custom study guides covering ML fundamentals, CNNs, RNNs, autoencoders, and transformers; all students achieved final grades of A

MIT New Engineering Education Transformation Winter 2024, Fall 2024, Spring 2024, Winter 2025

Undergraduate Lab Assistant (*5-person staff for 10-25 students under Dr. Gregory Long*)

- 16.632 Introduction to Autonomous Machines I and 2.S007 Design and Manufacturing I
- Mentored students in machining, Arduino programming, sensor integration, and mechanical design in preparation for a final autonomous robot competition; designed a web-based competition scoring system

Work Experience

Amazon Robotics Jun 2025 - Aug 2025

Robotics Intern Westborough, MA

- Organized a 5-intern team to implement scene-informed, transformer-based human trajectory prediction models trained on a self-collected, multi-modal dataset; deployed AWS infrastructure for real-time data collection and processing
- Built a React-based web UI for robot teleoperation and contributed to 2 ongoing FTR Innovation Lab robotics projects

Apple Inc. May 2024 - Aug 2024

Software Engineering Intern Cupertino, CA

- Created a 2-stage workflow automation tool in Django to source reference codebases, update configurations, generate code using Jinja, and manage Git and binary builds; supported 5+ CoreOS teams on the Apple Watch product line

Raytheon Intelligence & Space May 2022 - Aug 2022

Software Engineering Intern (Secret Security Clearance) El Segundo, CA

- Developed an automated Agile Metrics Tool to analyze, plot, and email reports of 200+ teams' execution data hosted on the Atlassian tool suite, including a web-based dashboard to visually represent and aggregate continuous integration build data

Leadership & Activities

Eta Kappa Nu (HKN), IEEE Honor Society May 2025 - Present

MIT Symphony Orchestra: Concertmaster, Stage Manager, Violinist Sep 2021 - May 2026

MIT E33 Lighting Productions: Chief Operating Officer, Production Manager, Technician Sep 2021 - May 2026

MIT Simmons Hall Front Desk: Desk Captain, Desk Worker Sep 2022 - May 2025